

UNIT II – ARITHMETIC AND LOGIC UNIT

Data Path Design: Components and Instruction Flow , ALU Design: Arithmetic and Logic Circuits,Integer Arithmetic: Booth's Algorithm, Overflow Detection Floating Point Arithmetic: IEEE 754, Microinstruction Format and Execution , FSM-based Control Design , Introduction to Verilog/VHDL

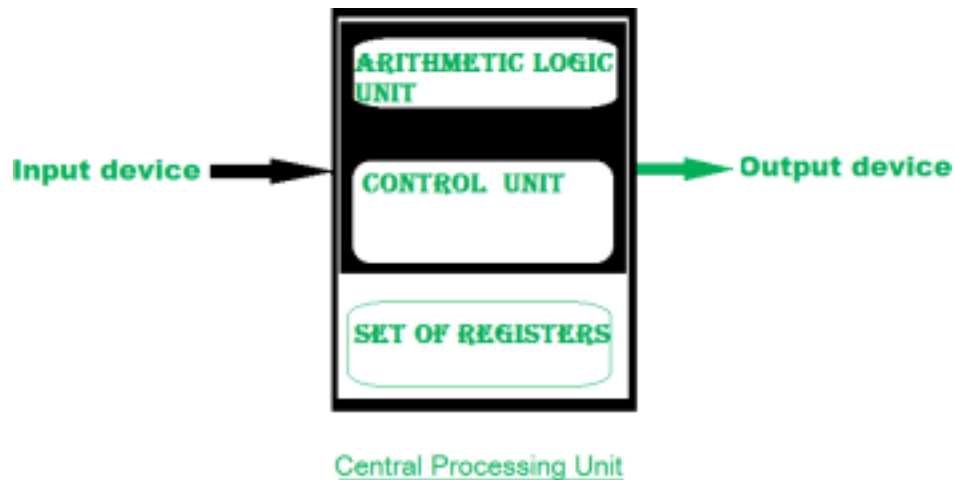
DATA PATH DESIGN:



What is ALU?

ALU is a digital circuit that provides arithmetic and logic operations. It is the fundamental building block of the central processing unit of a computer. A modern central processing unit(CPU) has a very powerful ALU and it is complex in design. In addition to ALU modern CPU contains a control unit and a set of registers. Most of the operations are performed by one or more ALUs, which load data from the input register. Registers are a small amount of

storage available to the CPU. These registers can be accessed very fast. The control unit tells ALU what operation to perform on the available data. After calculation/manipulation, the ALU stores the output in an output register.



The CPU can be divided into two sections: the **data section** and the **control section**. The *data section is also known as the data path*.

An **Arithmetic Logic Unit (ALU)** is a key component of the CPU responsible for performing arithmetic and logical operations. The collection of functional units like ALUs, registers, and buses that move data within the processor, together are known as Data Path, they execute instructions and manipulate data during processing tasks.

BUS

In early computers BUS were parallel electrical wires with multiple hardware connections. Therefore a bus is a communication system that transfers data between components inside a computer, or between computers. It includes hardware components like wires, optical fibers,

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etc and software, including communication protocols. The Registers, ALU, and the interconnecting BUS are collectively referred to as data paths.

Types of the bus

There are mainly three type of bus:-

1. **Address bus:** Transfers memory addresses from the processor to components like storage



and input/output devices. It's one-way communication.

2. **Data bus:** carries the data between the processor and other components. The data bus is bidirectional.

3. **Control bus:** carries control signals from the processor to other components. The control bus also carries the clock's pulses. The control bus is unidirectional.

The bus can be dedicated, i.e., it can be used for a single purpose or it can be multiplexed, i.e., it can be used for multiple purposes. When we would have different kinds of buses, different types of bus organizations will take place.

Registers

In Computer Architecture, the Registers are very fast computer memory which is used to execute programs and operations efficiently. But in that scenario, registers serve as gates, sending signals to various components to carry out little tasks. Register signals are directed by the control unit, which also operates the registers.

The following list of five registers for in-out signal data storage:

1. Program Counter

A program counter (PC) is a CPU register in the computer processor which has the address of the next instruction to be executed from memory. As each instruction gets fetched, the program counter increases its stored value by 1. It is a digital counter needed for faster execution of tasks as well as for tracking the current execution point.

Instruction Register

In computing, an instruction register (IR) is the part of a CPU's control unit that holds the instruction currently being executed or decoded. The instruction register specifically holds the instruction and provides it to the instruction decoder circuit.

3. Memory Address Register

The Memory Address Register (MAR) is the CPU register that either stores the memory address from which data will be fetched from the CPU, or the address to which data will be sent and stored. It is a temporary storage component in the CPU (central processing unit) that temporarily stores the address (location) of the data sent by the memory unit until the instruction for the particular data is executed.

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4. Memory Data Register

The memory data register (MDR) is the register in a computer's processor, or central processing unit, CPU, that stores the data being transferred to and from the

immediate



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access storage. Memory data register (MDR) is also known as memory buffer register (MBR).

5. General Purpose Register

General-purpose registers are used to store temporary data within the microprocessor . It is a multipurpose register. They can be used either by a programmer or by a user. What is

Data Path?

Suppose that the CPU needs to carry out any data processing action, such as copying data from memory to a register and vice versa, moving register content from one register to another, or adding two numbers in the ALU. Therefore, whenever a data processing action takes place in the CPU, the data involved for that operation follows a particular path, or data path.

Data paths are made up of various functional components, such as multipliers or arithmetic logic units. Data path is required to do data processing operations.

One Bus Organization

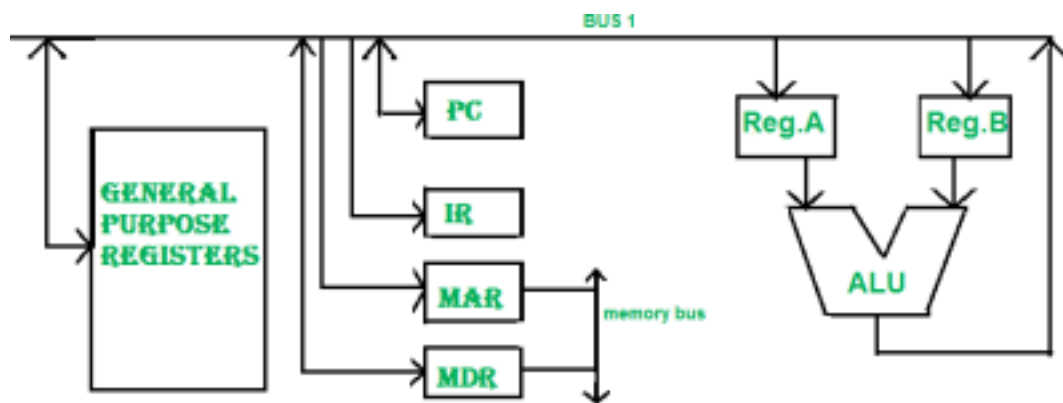


Fig. One Bus Organization

In one bus organization, a single bus is used for multiple purposes. A set of general-purpose registers, program counters, instruction registers, memory address registers (MAR), memory data registers (MDR) are connected with the single bus. Memory read/write can be done with MAR and MDR. The program counter points to the memory location from where the next instruction is to be fetched. Instruction register is that very register will hold the copy of the current instruction. In the case of one bus organization, at a time only one operand can be read from the bus.

As a result, if the requirement is to read two operands for the operation then the read operation needs to be carried twice. So that's why it is making the process a little longer. One of the advantages of one bus organization is that it is one of the simplest and also this is very

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cheap to implement. At the same time a disadvantage lies that it has only one bus and this "one bus" is accessed by all general-purpose registers, program counter, instruction register, MAR, MDR making each and every operation sequential. No one recommends this architecture nowadays.

Two Bus Organization

To overcome the disadvantage of one bus organization another architecture was developed

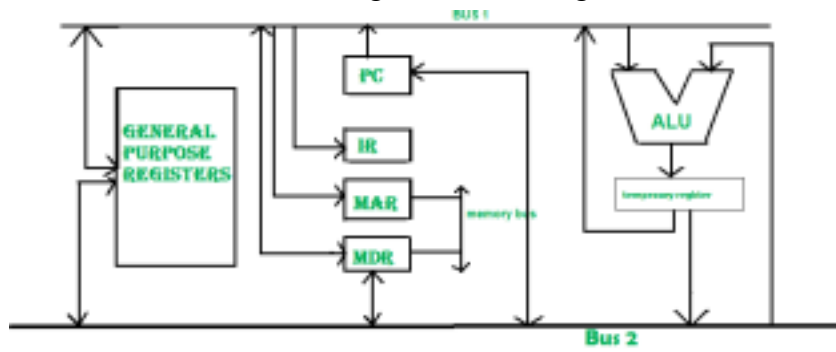


fig.TWO BUS ORGANISATION



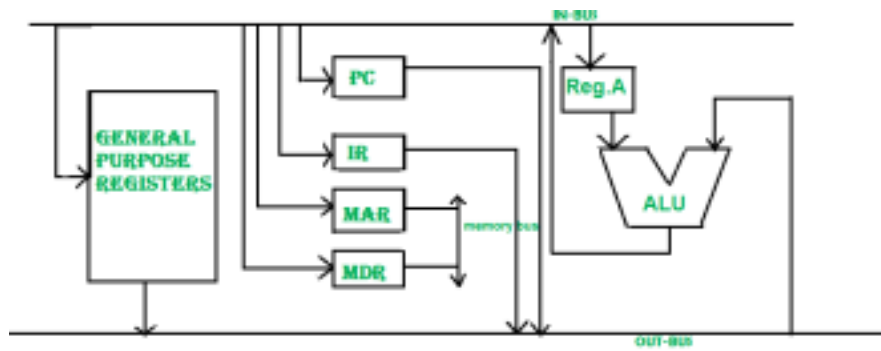


Fig. TWO BUS ORGANISATION

known as two bus organization. In two bus organizations, there are two buses. The general-purpose register can read/write from both the buses. In this case, two operands can be fetched at the same time because of the two buses. One bus fetch operand for ALU and another bus fetch for register. The situation arises when both buses are busy fetching operands, the output can be stored in a temporary register and when the buses are free, the particular output can be dumped on the buses.

There are two versions of two bus organizations, i.e., in-bus and out-bus. From in-bus, the general-purpose register can read data and to the out bus, the general-purpose registers can write data. Here buses get dedicated.

Three Bus Organization

In three bus organizations we have three buses, OUT bus1, OUT bus2, and an IN bus. From the out buses, we can get the operand which can come from the general-purpose register and evaluated in ALU and the output is dropped on In Bus so it can be sent to respective registers. This implementation is a bit complex but faster in nature because in parallel two operands



can flow into ALU and out of ALU. It was developed to overcome the busy waiting problem of two bus organizations. In this structure after execution, the output can be dropped on the bus without waiting because of the presence of an extra bus. The structure is given below in the figure.

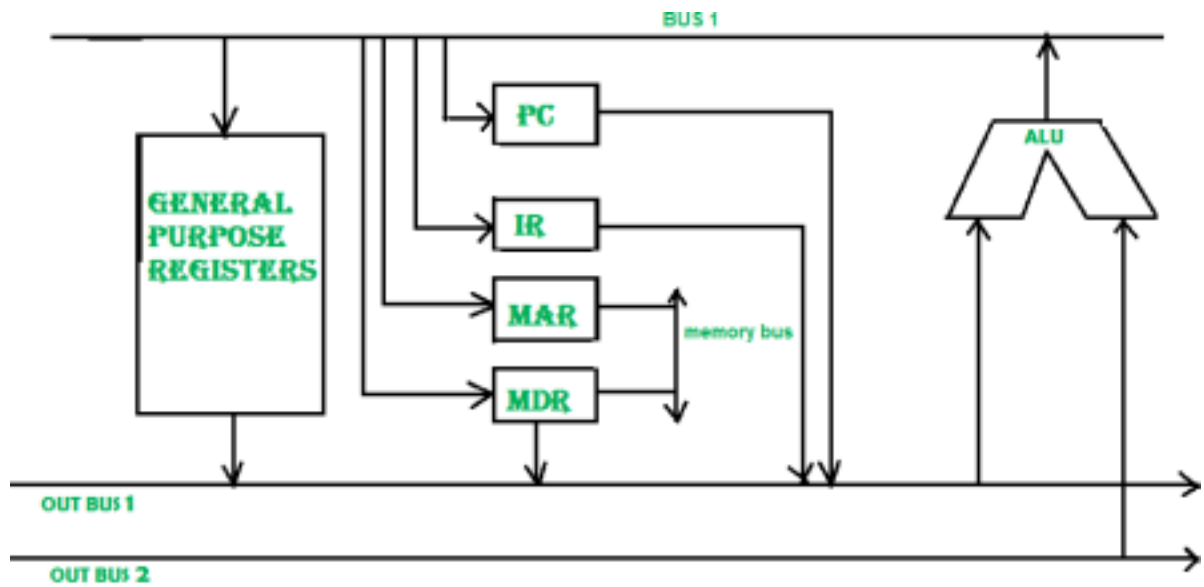


Fig. THREE BUS ORGANISATION

The main advantages of multiple bus organizations over the single bus are as given below.

1. Increase in size of the registers.
2. Reduction in the number of cycles for execution.
3. Increases the speed of execution or we can say faster execution.

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Computer Arithmetic

Introduction:

Data is manipulated by using the arithmetic instructions in digital computers. Data is manipulated to produce results necessary to give solution for the computation problems. The Addition, subtraction, multiplication and division are the four basic arithmetic operations. If we want then we can derive other operations by using these four operations.

To execute arithmetic operations there is a separate section called arithmetic processing unit in central processing unit. The arithmetic instructions are performed generally on binary or decimal data. Fixed-point numbers are used to represent integers or fractions. We can have signed or unsigned negative numbers. Fixed-point addition is the simplest arithmetic operation.

If we want to solve a problem then we use a sequence of well-defined steps. These steps are collectively called algorithm. To solve various problems we give algorithms.

In order to solve the computational problems, arithmetic instructions are used in digital computers that manipulate data. These instructions perform arithmetic calculations.

And these instructions perform a great activity in processing data in a digital computer. As we already stated that with the four basic arithmetic operations addition, subtraction, multiplication and division, it is possible to derive other arithmetic operations and solve scientific problems by means of numerical analysis methods.

A processor has an arithmetic processor(as a sub part of it) that executes arithmetic operations. The data type, assumed to reside in processor, registers during the execution of an arithmetic instruction. Negative numbers may be in a signed magnitude or signed complement representation. There are three ways of representing negative fixed point - binary numbers signed magnitude, signed 1's complement or signed 2's complement. Most computers use the signed magnitude representation for the mantissa.



Addition and Subtraction :

Addition and Subtraction with Signed –Magnitude Data

We designate the magnitude of the two numbers by A and B. Where the signed numbers are added or subtracted, we find that there are eight different conditions to consider, depending on the sign of the numbers and the operation performed. These conditions are listed in the first column of Table 4.1. The other columns in the table show the actual operation to be performed with the magnitude of the numbers. The last column is needed to present a negative zero. In other words, when two equal numbers are subtracted, the result should be +0 not -0.

The algorithms for addition and subtraction are derived from the table and can be stated as follows (the words parentheses should be used for the subtraction algorithm)

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Floating Point Arithmetic operation



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Algorithm for Floating Point Addition and Subtraction









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Overflow in Arithmetic Addition

Conditions of Overflow

Overflow takes place when the result of the calculations exceeds the range of the number to be represented in a fixed number of bits with the help of 2's complement format. Specifically, overflow happens under the following conditions:



1) Addition of Two Positives

When adding two positive numbers and the amount is more than the maximum amount that an N-bit system can hold, then overflow takes place. In case it becomes the most significant bit is set to one signifying that the result is negative and this is wrong.

2) Addition of two negative number

Overflow is obtained in the subtraction, when two negative numbers are summed up and with the result, which is placed below the minimum representable value. Here the MSB becomes 0 which means positive result which is also wrong.

3) Variety of Carry in and Carry Out bits

This form of overflow is identified by comparing the carry-in and the carry-out relating to the MSB in an addition. However, if the two bits are at different state, that is C-in is not equal to C-out, then an overflow has taken place. This condition enables a determination of when the

result cannot be represented using the available N-bit space.

Overflow Detection

Overflow occurs when:

1. Two negative numbers are added and an answer comes positive or
2. Two positive numbers are added and an answer comes as negative.

IEEE Standard 754 Floating Point Numbers

IEEE 754 has 3 basic components:

1. **The Sign of Mantissa** - This is as simple as the name. 0 represents a positive number while 1 represents a negative number.
2. **The Biased exponent** - The exponent field needs to represent both positive and negative exponents. A bias is added to the actual exponent in order to get the stored exponent.
3. **The Normalised Mantissa** - The mantissa is part of a number in scientific notation or a floating-point number, consisting of its significant digits. Here we have only 2 digits, i.e. 0 and 1. So a normalised mantissa is one with only one 1 to the left of the decimal.

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**IEEE 754 numbers are divided into two based on the above three components:
single precision and double precision.**





TYPES	SIGN	BIASED EXPONENT	NORMALISED MANTISA	BIAS
Single precision	1(31st bit)	8(30-23)	23(22-0)	127
Double precision	1(63rd bit)	11(62-52)	52(51-0)	1023